

Vlad Voropaev

Computer Vision Engineer | Industrial Video Analytics | Detection, Tracking & Pose

Dubai, UAE | thevladvoropaev@gmail.com | linkedin.com/in/thevladvoropaev | github.com/voropaev

SUMMARY

Computer Vision Engineer with 4+ years across industrial video analytics and applied machine learning. Built multi-camera safety systems from dataset design and held-out evaluation through tracking, pose logic, RTSP integration, field iteration, and paid customization. Former Cyclop co-founder and Deputy CEO; first author and oral presenter of an IEEE Computer Vision paper.

TECHNICAL SKILLS

Computer Vision: Object detection, object tracking, multi-camera video analytics, pose estimation, PPE and zone analytics, image preprocessing, OCR

ML & Libraries: Python, PyTorch, OpenCV, Ultralytics YOLO, TensorFlow, ByteTrack, Deep SORT, NumPy

Systems & Evaluation: RTSP, FastAPI, Docker, PostgreSQL, GPU inference, dataset design and annotation, held-out evaluation, error analysis

EXPERIENCE

LLC Cyclop — Co-founder, Deputy CEO & Applied AI Engineer (Computer Vision) Oct 2022 – Jul 2026
Industrial Computer Vision, client customization, pilot integration, product definition, and grant-funded R&D

- Owned a six-camera crane-operator safety system end to end: built a **1,163-image / 7,499-object dataset**, trained YOLOv8m, and reached **0.932 precision, 0.878 recall, 0.920 mAP@0.50, and 0.707 mAP@0.50:0.95** on a held-out test set
- Implemented active-view selection, ByteTrack identities, pose-based wrist-to-pendant association, helmet checks, temporal voting, and saved event evidence; replaced the initial IoU tracker and iterated on a white-helmet field failure. The work became a paid client customization
- Co-developed modular RTSP analytics for PPE, danger zones, tracking/counting, equipment use, and worker activity; offline PPE validation reached **0.906 mAP@0.50 and 0.772 mAP@0.50:0.95** on 579 images / 1,748 objects
- Reviewed **17,696 CCTV frames from eight camera feeds** and curated **1,156 relevant frames** for a multi-camera industrial pilot; designed view-specific zones to reduce long-range and duplicate detections across overlapping views
- Built a worker-activity analytics prototype for low-resolution CCTV using detection, tracking, pose, fixed work zones, temporal stabilization, repeat-alert suppression, reason labels, and image/video evidence
- Built NeuroQuest, an end-to-end RTSP-to-comic system combining detection, tracking, face/action recognition, scene interpretation, language and image generation, and PDF assembly; served as first author, led the paper's writing, and presented it at ACDSA 2024 ([IEEE Xplore 10467698](#))

Shtrih-M — Computer Vision Engineer Intern Feb 2022 – Jun 2022

- Developed and visually evaluated a Python/OpenCV preprocessing workflow across **1,147 raw GRBG Bayer images** captured under varied lighting, glare, backgrounds, and packaging conditions; combined demosaicing, highlight suppression, CLAHE, and adaptive gamma correction
- Documented seven image-processing functions, before/after comparisons, RGB histograms, adaptive decision logic, and implementation details in a **64-page technical report**

Sberbank — Data Scientist Intern — Research & Innovation Sep 2021 – Nov 2021

- Worked on a loan-delinquency prediction task in Sberbank's Research and Innovation Directorate; contributed to developing and evaluating predictive-modeling approaches using ROC-AUC

EDUCATION & PUBLICATIONS

Bauman Moscow State Technical University Sep 2022 – Jul 2024
M.S., Information Systems and Technologies — with honors

Bauman Moscow State Technical University Sep 2018 – Jul 2022
B.S., Information Systems and Technologies — with honors

Publications: first-author IEEE paper on NeuroQuest (ACDSA 2024); single-author classical Computer Vision article on polygonal-zone detection (Vestnik Nauki, 2023)

Languages: Russian — native; English — professional working proficiency